

Cables and Connectors

Course Code: CSC 3116

Course Title: Computer Networks



Dept. of Computer Science
Faculty of Science and Technology

Lecturer No:	Lab 4	Week No:	2	Semester:	
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Lecture Outline



1. Different types of devices
2. Ethernet Cabling
3. Terminal Emulation Software
4. Remote Access

Cisco Switch



Fig. A 2960 series cisco switch

CLI access options

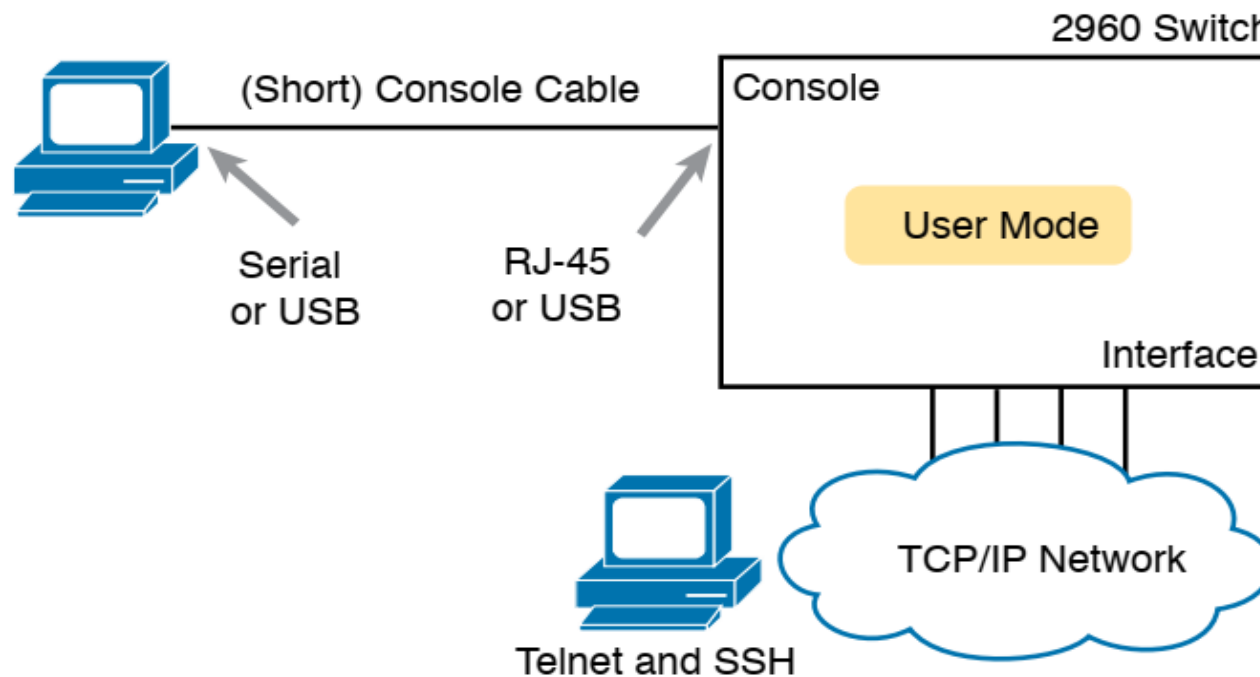


Fig. CLI access options

Console Connection to Switch

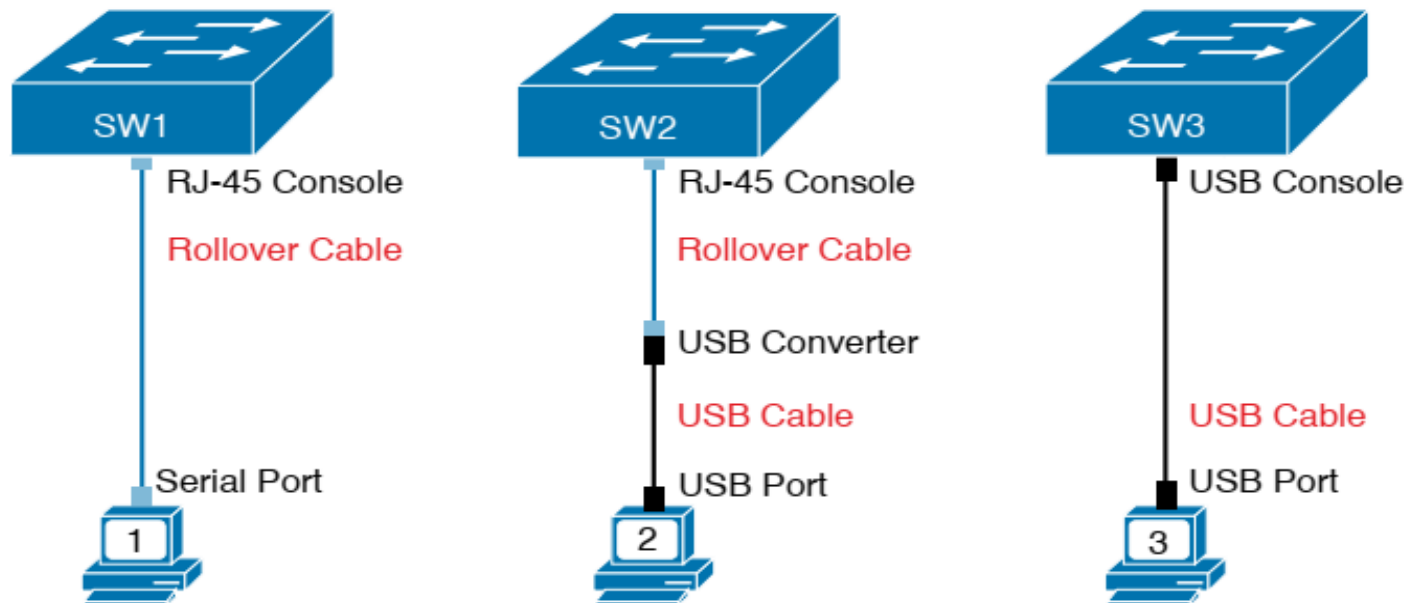
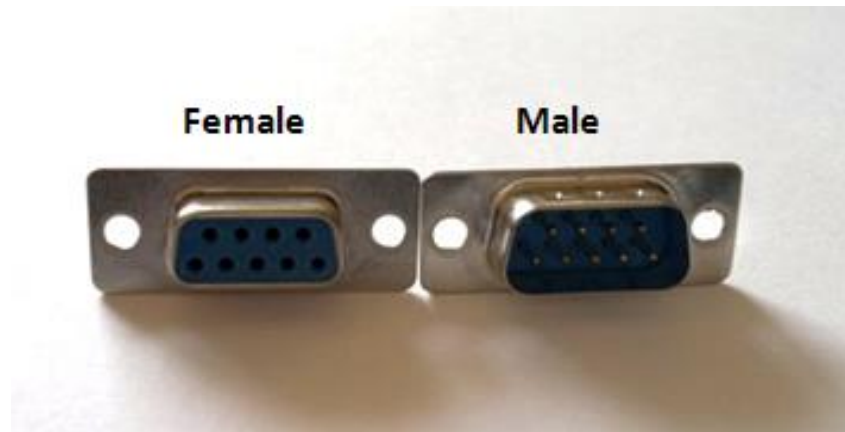


Fig. Console Connection to a Switch

2960-XR series supports both the older RJ-45 console port and a USB console port

USB console port is Not rectangular ,rather it is Mini-B port

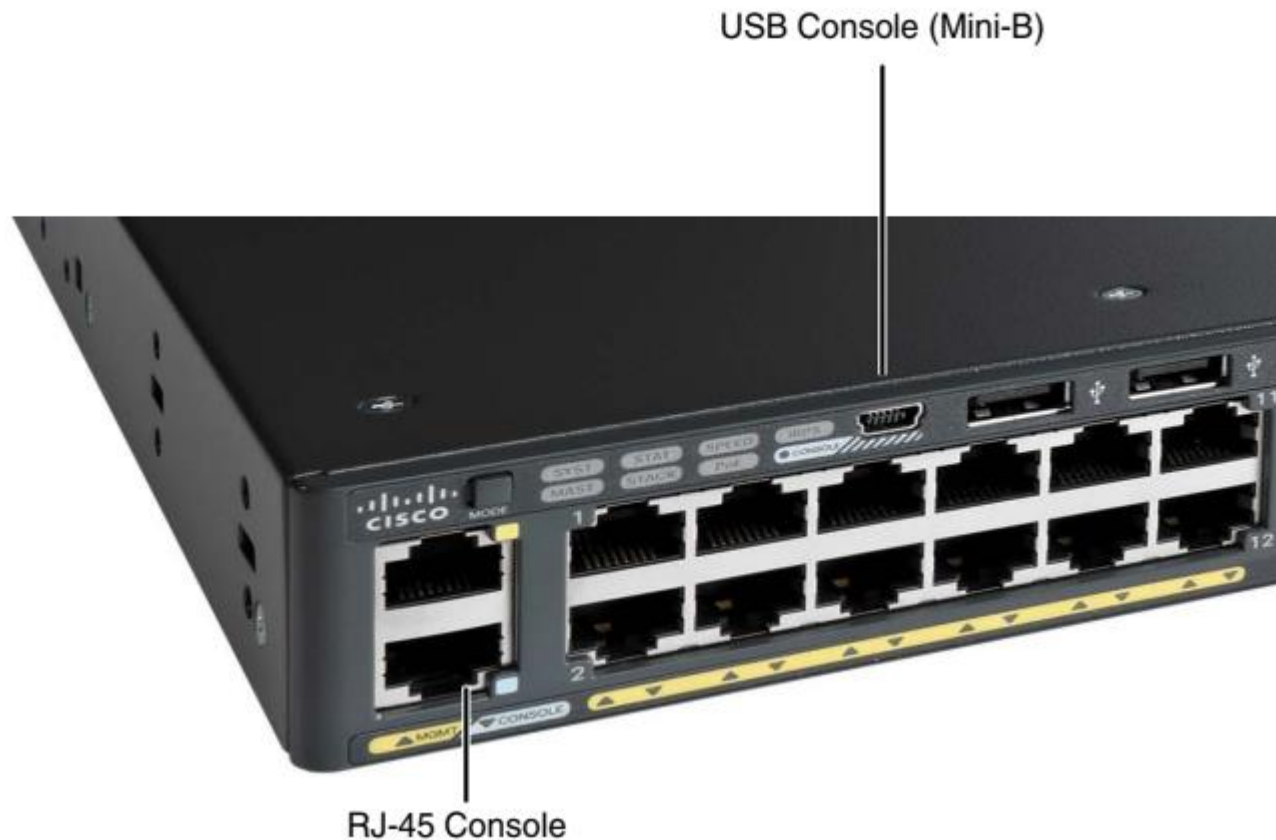
DB9 Connector



Almost obsolete
Replaced by USB

Fig. DB9 Connector (For serial port of PC)

USB and RJ-45 ConSOLE



Terminal Emulation Software



To interact with Switch/Routers

Examples

- SecureCRT: Supports serial, Telnet and SSH → Not free
- PuTTY: Supports serial, Telnet and SSH → free
- TeraTerm Pro: Supports serial, Telnet and SSH → free
- Windows Telnet (integrated to Windows OS) → Supports Telnet only
- Windows HyperTerminal → Supports Telnet and Serial, NOT SSH

Remote Access

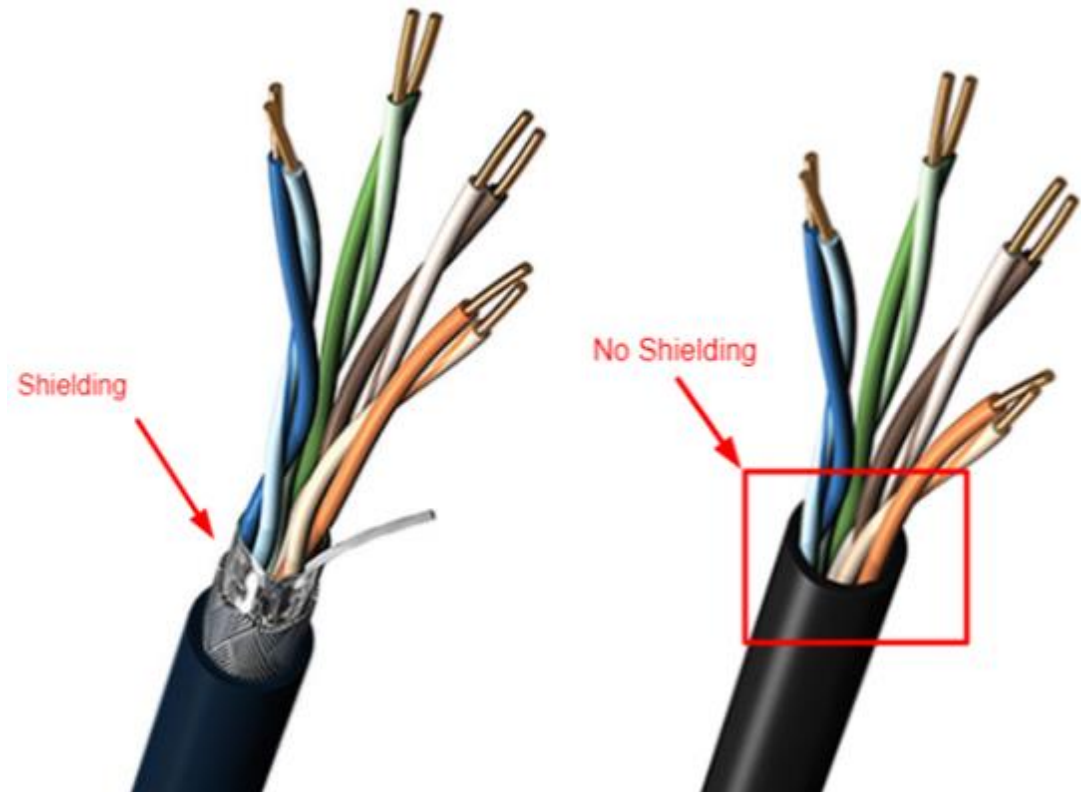


- Telnet and SSH allows remote access to Switch/Router
- Must need password to access the device
- Telnet is not secure
 - No encryption
- SSH is secure
 - Encryption is used

Ethernet cabling



- Unshielded Twisted Pair (UTP)
 - Electromagnetic interference
 - Less data rate
 - cheap
- Shielded Twisted Pair (STP)
 - Less interference
 - Higher data rate
 - costly



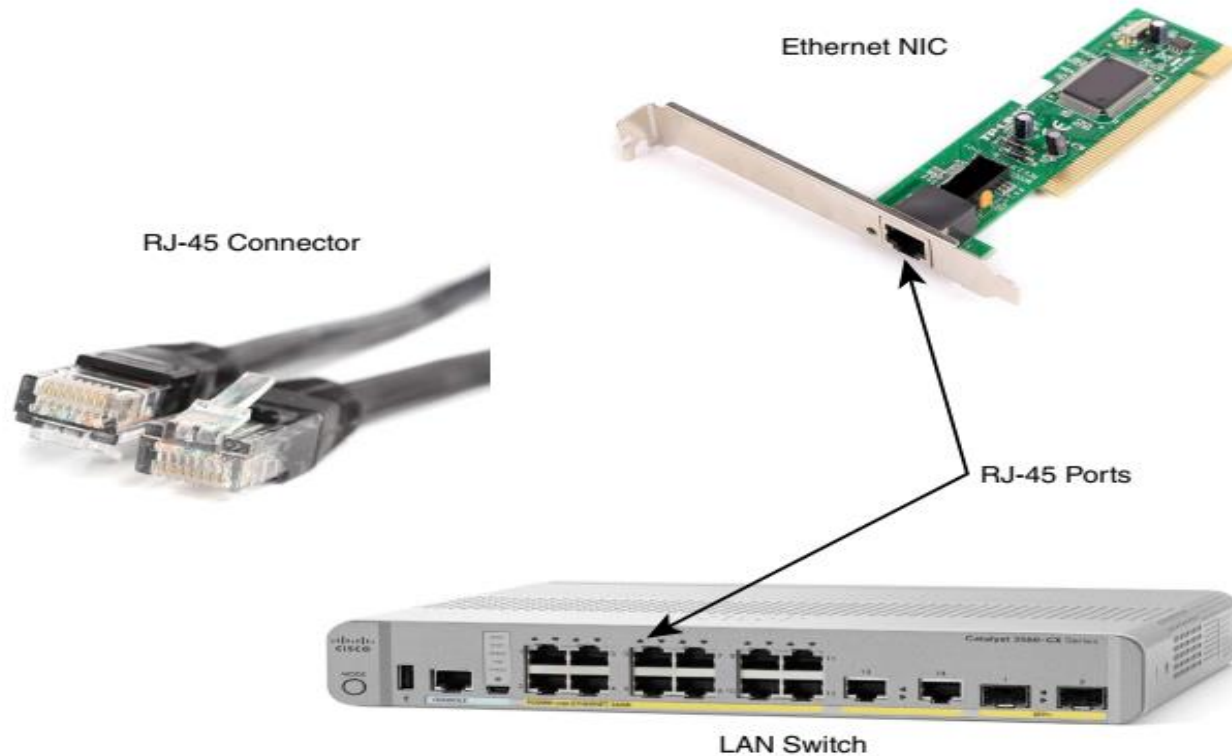
Ethernet cabling (contd...)



Table I Different categories of cable

Category	Speed (Mbps)	Common use
Cat 1	< 1	Analog voice
Cat 2	4	ARCNET
Cat 3	10	10baseT Ethernet
Cat 4	16	Token Ring
Cat 5	100	100baseT Ethernet
Cat 5e	1000	1000baseT Ethernet
Cat 6	1000	1000baseT Ethernet

Ethernet cabling (contd...)



Ethernet cabling (contd...)



- Straight-Through cable
 - PC to Switch or Hub
 - Router to Switch or Hub
- Crossover cable
 - PC to PC
 - Switch to Switch
 - Hub to Hub
 - Switch to Hub
 - Router to Router
 - Router to PC
- Rollover
 - PC to Console

Crossover cable: If the endpoints transmit on the same pin pair

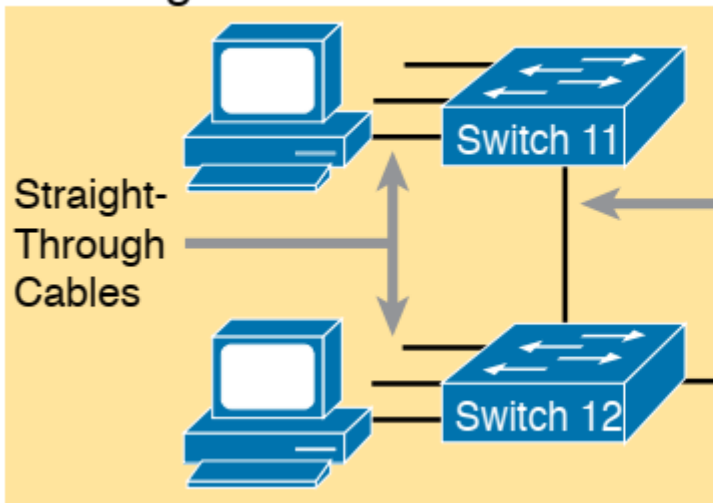
Straight-through cable: If the endpoints transmit on different pin pairs

Transmits on Pins 1,2	Transmits on Pins 3,6
PC NICs	Hubs
Routers	Switches
Wireless access point (Ethernet interface)	—

Ethernet cabling (contd...)

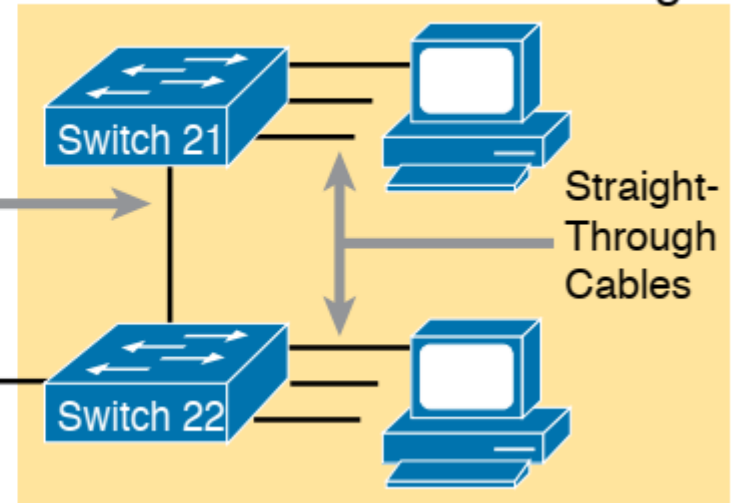


Building 1



Crossover
Cables

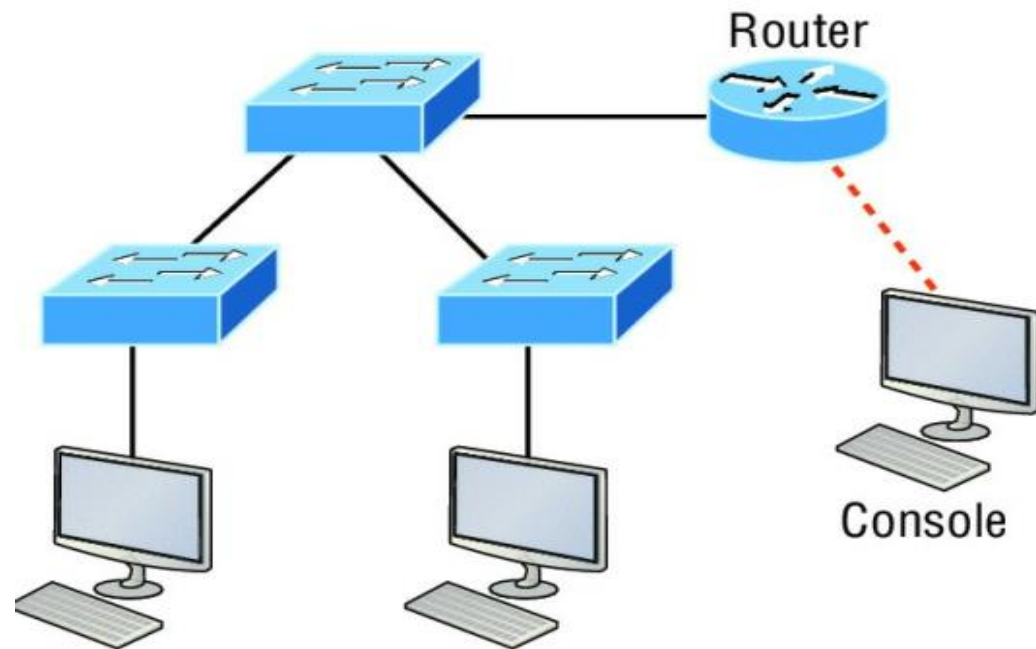
Building 2



Ethernet cabling (contd...)



Identify cables

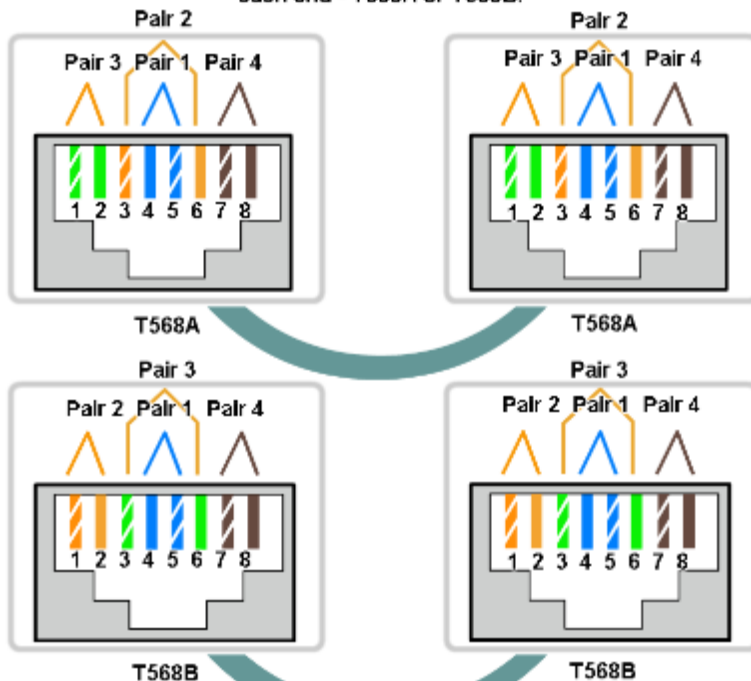


Ethernet cabling (contd...)



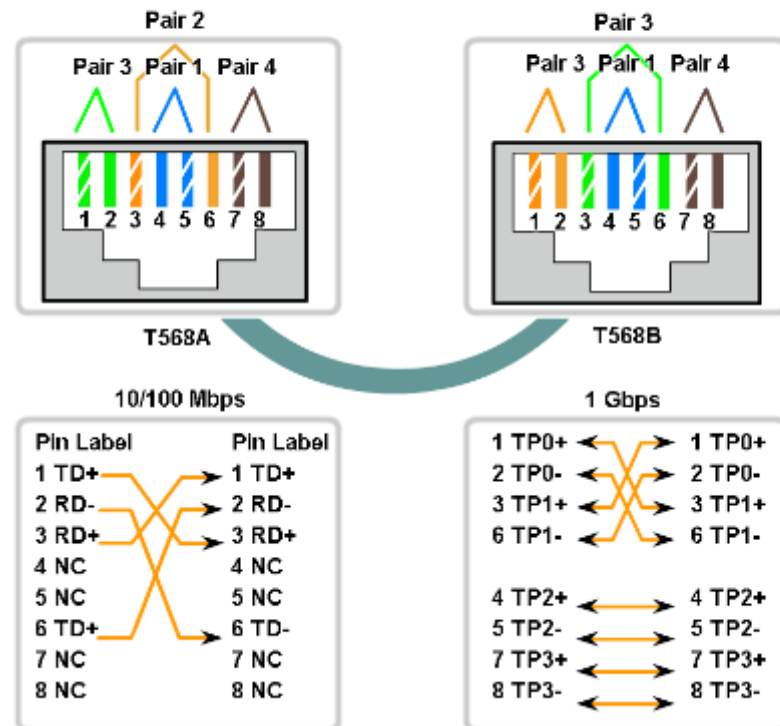
Straight-Through Cable

Straight-through cables have the same termination at each end - T568A or T568B.



Crossover Cable

Crossover cables have a T568A termination at one end and a T568B termination at the other end.



Transmit pins at each end connect to the receive pins at the other end.



- Suppose that we have three networks BBA, CSE, and EEE with IP requirements of 100, 350 and 80.

If You are given an IP block 10.10.0.0, allocate IPs performing subnetting.

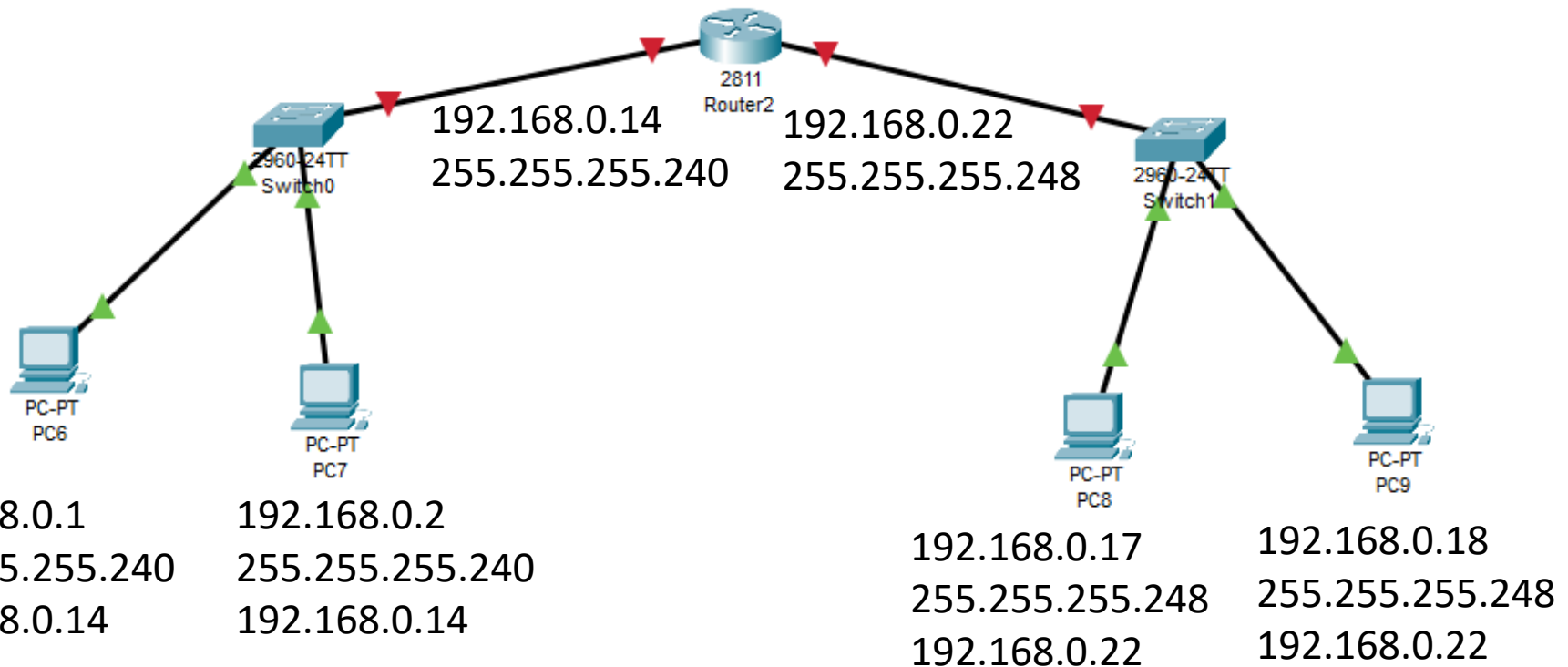
- Suppose that we have five networks A, B, C, D, and E with IP requirements of 10, 30, 90, 200, and 80.

If You are given an IP block 172.16.0.0, allocate IPs performing subnetting.

- Consider your ID: AB-CDEFG-H, Suppose that we have four networks A, B, C, and D with IP requirements of $100 \times C$, $10 \times D$, $50 \times A$, and $20 \times G$.

If You are given an IP block 192.168.0.0, allocate IPs performing subnetting.

➤ Suppose that we have two networks CSE and EEE with IP requirements of 10 and 8. If You are given an IP block 192.168.0.0, allocate IPs performing subnetting.



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192.168.0.0/16

CSE → 10 host → 4bit → 192.168.0.0-15/28 → 255.255.255.240

Net address: 192.168.0.0/28

BC address: 192.168.0.15/28

Gateway: 192.168.0.14/28

EEE → 8 host → 3bit → 192.168.0.16-23/29 → 255.255.255.248

Net address: 192.168.0.16/29

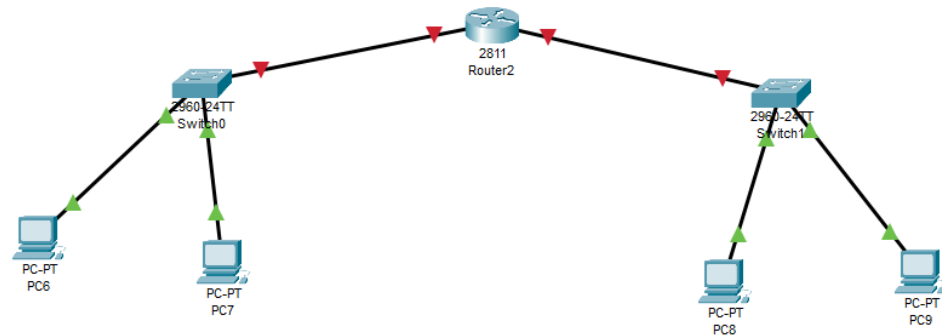
BC address: 192.168.0.23/29

Gateway: 192.168.0.22/29

RN → 4 IP → 2bit →

192.168.0.24~27/30

255.255.255.252





References

1. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
2. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.
3. Cisco IOS Configuration Fundamentals Command Reference.
<http://www.cisco.com>



Books

1. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
2. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.